

Task 5 – RAG Document Q&A: Output Screenshots

1. Indexing check (GET /documents/count)

The PDF was split into chunks, embedded and stored in the vector database, with 102 chunks indexed.

The screenshot displays a REST client interface with the following sections:

- GET /documents/count Document Count**: The endpoint and method.
- Parameters**: A section with a "Cancel" button and the text "No parameters".
- Execute**: A blue button to execute the request.
- Clear**: A button to clear the request.
- Responses**: A section containing:
 - Curl**: A code block with the command:

```
curl -X 'GET' \
'http://127.0.0.1:8000/documents/count' \
-H 'accept: */*'
```
 - Request URL**: A text field containing `http://127.0.0.1:8000/documents/count`.
 - Server response**: A section with a table showing the response details.

Code	Details
200	Response body<pre>{ "indexed_chunks": 102 }</pre> Download Response headers<pre>content-length: 22 content-type: application/json date: Fri, 09 Oct 2026 08:04:21 GMT server: uvicorn</pre>

2. Retrieval before cleaning (POST /ask)

The AR vs VR question returned relevant chunks (distance ~0.33–0.37) and a grounded answer, but the stored text was noisy with stray \n characters.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "question": "What is the difference between Augmented Reality and Virtual Reality?",
    "top_k": 4
  }'
```

Request URL

http://127.0.0.1:8000/ask

Server response

Code

Details

200

Response body

```
"question": "What is the difference between Augmented Reality and Virtual Reality?",
"answer": "Based on the provided documents, the differences are:\n\n* **Augmented Reality (AR):** Does not replace the user's surroundings, but rather enhances the physical world by overlaying digital elements (such as three-dimensional models, images, text, or animations) onto real-world environments to enrich them with additional contextual information (Source: *ar_vr_medical_training.pdf*, pages 2 and 7).\n* **Virtual Reality (VR):** Creates a completely simulated, computer-generated digital environment that replaces the real-world surroundings, and users interact with it using specialized hardware (Source: *ar_vr_medical_training.pdf*, page 2).",
"sources": [
  {
    "document": "ar_vr_medical_training.pdf",
    "page": 2,
    "chunk_id": "chunk_5",
    "distance": 0.3304,
    "text": "display\n\nvirtual\n\norgans\n\nnor\n\nblood\n\nvessels\n\nnduring\n\nprocedures,\n\nhelping\n\nsurgeons\n\nperform\n\noperations\n\nwith\n\nhigher\n\nprecision.\n\nCombining\n\nreal\n\nand\n\ndigital\n\ninformation\n\nin\n\nreal\n\ntime,\n\nAR\n\nenhances\n\ninstitutional\n\nawareness\n\nand\n\nimproves\n\ndecision-making\n\nduring\n\nmedical\n\ntraining\n\nand\n\nclinical\n\npractice.\n\n1.2\n\nOverview\n\nof\n\nVirtual\n\nReality\n\n(VR)\n\nVirtual\n\nReality\n\nis\n\nan\n\nimmersive\n\ntechnology\n\nthat\n\ncreates\n\nan\n\ncompletely\n\nsimulated\n\ndigital\n\nenvironment\n\nin\n\nwhich\n\nusers\n\ncan\n\ninteract\n\nusing\n\nspecialized\n\nhardware\n\nsuch\n\nas\n\nhead-mounted\n\ndisplays,\n\nmotion\n\ncontrollers,\n\nand\n\ntracking\n\nsensors.\n\nVR\n\nreplaces\n\nthe\n\nreal-world\n\nsurroundings\n\nwith\n\nan\n\ncomputer-generated\n\nenvironment.",
  },
  {
    "document": "ar_vr_medical_training.pdf",
    "page": 1,
    "chunk_id": "chunk_0",
    "distance": 0.371,
    "text": "1.\n\nINTRODUCTION\n\nAugmented\n\nReality\n\n(AR)\n\nand\n\nVirtual\n\nReality\n\n(VR)\n\nare\n\nadvanced\n\nimmersive\n\ntechnologies\n\nthat\n\nare\n\ntransforming\n\nthe\n\nway\n\nwe\n\nacquire\n\nknowledge\n\nis\n\nundelivered\n\nand\n\npractical\n\nskills\n\nare\n\ndeveloped\n\nacross\n\nvarious\n\nindustries,\n\nparticularly\n\nin\n\nhealthcare\n\nand\n\nmedical\n\neducation.\n\nThese\n\ntechnologies\n\nprovide\n\ninteractive\n\ndigital\n\nenvironments\n\nthat\n\nenhance\n\nvisualization\n\nand\n\nenable\n\nusers\n\nto\n\nexperience\n\nsimulated\n\nreal-world\n\nscenarios.\n\nIn\n\nthe\n\nfield\n\nof\n\nmedical\n\ntraining\n\nand\n\nsurgical\n\nsimulation,\n\nAR\n\nand\n\nVR\n\nhave\n\ninnovative\n\nsolutions\n\nthat\n\nallow\n\nmedical\n\nstudents\n\nand\n\nprofessionals\n\nto\n\npractice\n\nprocedures\n\nsafely\n\nwithout\n\nexposing\n\nreal\n\nto\n\npotential"
```

Response headers

```
content-length: 4977
content-type: application/json
date: Fri, 09 Oct 2026 08:06:20 GMT
server: uvicorn
```

Responses

Code

Description

Links

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "question": "What is the difference between Augmented Reality and Virtual Reality?",
    "top_k": 4
  }'
```

Request URL

http://127.0.0.1:8000/ask

Server response

Code

Details

200

Response body

```
to\n\npotential"
},
{
  "document": "ar_vr_medical_training.pdf",
  "page": 7,
  "chunk_id": "chunk_24",
  "distance": 0.373,
  "text": "3.\n\nFUNDAMENTALS\n\nof\n\nAR\n\nand\n\nVR\n\nTECHNOLOGY\n\nAugmented\n\nReality\n\n(AR)\n\nand\n\nVirtual\n\nReality\n\n(VR)\n\nare\n\nimmersive\n\ntechnologies\n\nthat\n\nintegrate\n\ncomputer\n\ngraphics,\n\nsensor\n\nsystems,\n\ntracking\n\nalgorithms,\n\nand\n\ninteractive\n\nsoftware\n\nto\n\nenhance\n\nor\n\nsimulate\n\nreal-world\n\nenvironments.\n\nIn\n\nmedical\n\ntraining\n\nand\n\nsurgical\n\nsimulation,\n\nthese\n\ntechnologies\n\nform\n\nthe\n\nfoundation\n\nfor\n\ncreating\n\nrealistic\n\nand\n\ninteractive\n\nlearning\n\nplatforms.\n\nUnderstanding\n\nthe\n\nfundamental\n\nconcepts,\n\noperational\n\nprinciples,\n\nand\n\ntechnical\n\ncomponents\n\nof\n\nAR\n\nand\n\nVR\n\nsystems\n\nis\n\nessential\n\nfor\n\nanalyzing\n\ntheir\n\neffectiveness\n\nin\n\nhealthcare\n\neducation.\n\n3.1\n\nAugmented\n\nReality\n\nconcepts\n\nAugmented\n\nReality\n\nis\n\nan\n\nimmersive\n\ntechnology\n\nthat\n\nenhances\n\nthe\n\nphysical\n\nworld\n\nby\n\noverlying\n\ndigital\n\nelements\n\nsuch\n\nas\n\nthree-dimensional\n\nmodels,\n\nimages,\n\ntext,\n\nor\n\nanimations\n\nonto\n\nreal-world\n\nenvironments.\n\nUnlike\n\nVirtual\n\nReality,\n\nAR\n\ndoes\n\nnot\n\nreplace\n\nthe\n\nuser's\n\nsurroundings\n\nbut\n\nenriches\n\nthe\n\nwith\n\nadditional\n\ncontextual\n\ninformation.\n\nAR\n\nsystems\n\nuse\n\ncameras\n\nand\n\nsensors\n\nto\n\ndetect\n\nsurfaces,\n\nrecognize\n\nobjects,\n\nand\n\ndetermine\n\nspatial\n\npositioning\n\nthe\n\nsystem\n\nthen\n\naligns\n\nvirtual\n\nobjects\n\naccurately\n\nwithin\n\nthe\n\nuser's\n\nfield\n\nof\n\nview.\n\nIn\n\nmedical\n\napplications,\n\nAR\n\nenables"
```

Response headers

```
content-length: 4977
content-type: application/json
date: Fri, 09 Oct 2026 08:06:20 GMT
server: uvicorn
```

Responses

Code

Description

Links

3. Retrieval after cleaning (POST /ask)

After text cleaning, chunks are readable and the answer correctly contrasts AR (enhances the real world) with VR (replaces it), citing document and page.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "question": "What is the difference between Augmented Reality and Virtual Reality?",
    "top_k": 4
  }'
```

Request URL

http://127.0.0.1:8000/ask

Server response

Code

Details

200

Response body

```
{
  "question": "What is the difference between Augmented Reality and Virtual Reality?",
  "answer": "Based on the provided documents:\n\n* **Augmented Reality (AR)** enhances the real world by combining real and digital information in real time (such as displaying virtual organs or blood vessels during procedures) (Source: ar_vr_medical_training.pdf, page 2).\n* **Virtual Reality (VR)** creates a completely simulated, computer-generated digital environment that replaces the real-world surroundings, allowing users to interact with it using specialized hardware like head-mounted displays, motion controllers, and tracking sensors (Source: ar_vr_medical_training.pdf, page 2).",
  "sources": [
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 2,
      "chunk_id": "chunk_4",
      "distance": 0.3401,
      "text": "display virtual organs or blood vessels during procedures, helping surgeons perform operations with higher precision. By combining real and digital information in real time, AR enhances situational awareness and improves decision-making during medical training and clinical practice. 1.2 Overview of Virtual Reality (VR) Virtual Reality is an immersive technology that creates a completely simulated digital environment in which users can interact using specialized hardware such as head-mounted displays, motion controllers, and tracking sensors. VR replaces the real-world surroundings with a computer-generated environment, allowing users to experience realistic simulations of physical spaces and activities"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 1,
      "chunk_id": "chunk_0",
      "distance": 0.374,
      "text": "1. INTRODUCTION Augmented Reality (AR) and Virtual Reality (VR) are advanced immersive technologies that are transforming the way knowledge is delivered and practical skills are developed across various industries, particularly in healthcare and medical education. These technologies provide interactive digital environments that enhance visualization and enable users to experience simulated real-world scenarios. In the field of medical training and surgical simulation, AR and VR offer innovative solutions that allow medical students and professionals to practice procedures safely without exposing real patients to potential risks"
    },
    {
      "document": "ar_vr_medical_training.pdf"
    }
  ]
}
```

Response headers

```
content-length: 3466
content-type: application/json
date: Fri, 09 Oct 2026 08:34:13 GMT
server: uvicorn
```

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "question": "What is the difference between Augmented Reality and Virtual Reality?",
    "top_k": 4
  }'
```

Request URL

http://127.0.0.1:8000/ask

Server response

Code

Details

200

Response body

```
{
  "chunk_id": "chunk_0",
  "distance": 0.374,
  "text": "1. INTRODUCTION Augmented Reality (AR) and Virtual Reality (VR) are advanced immersive technologies that are transforming the way knowledge is delivered and practical skills are developed across various industries, particularly in healthcare and medical education. These technologies provide interactive digital environments that enhance visualization and enable users to experience simulated real-world scenarios. In the field of medical training and surgical simulation, AR and VR offer innovative solutions that allow medical students and professionals to practice procedures safely without exposing real patients to potential risks"
},
{
  "document": "ar_vr_medical_training.pdf",
  "page": 7,
  "chunk_id": "chunk_20",
  "distance": 0.3937,
  "text": "3. FUNDAMENTALS OF AR AND VR TECHNOLOGY Augmented Reality (AR) and Virtual Reality (VR) are immersive technologies that integrate computer graphics, sensor systems, tracking algorithms, and interactive software to enhance or simulate real-world environments. In medical training and surgical simulation, these technologies form the foundation for creating realistic and interactive learning platforms. Understanding the fundamental concepts, operational principles, and technical components of AR and VR systems is essential for analyzing their effectiveness in healthcare education"
},
{
  "document": "ar_vr_medical_training.pdf",
  "page": 6,
  "chunk_id": "chunk_19",
  "distance": 0.4084,
  "text": ". Augmented Reality and Virtual Reality technologies offer a potential solution by creating interactive simulation systems that replicate real surgical environments. These systems enhance skill acquisition, improve confidence, and ensure patient safety. Therefore, the motivation behind this study is to analyze how AR/VR-based medical training systems can overcome existing limitations and contribute to improved healthcare education and outcomes. 6"
}
}
```

Response headers

```
content-length: 3466
content-type: application/json
date: Fri, 09 Oct 2026 08:34:13 GMT
server: uvicorn
```

4. Hallucination check (out-of-scope question)

For an unrelated question (population of India), retrieval distances were high (~0.90+) and the model refused rather than inventing an answer.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "question": "What is the population of India according to this report?",
    "top_k": 4
  }'
```

Request URL

http://127.0.0.1:8000/ask

Server response

Code Details

200

Response body

```
{
  "question": "What is the population of India according to this report?",
  "answer": "The provided documents do not contain enough information to answer this question.",
  "sources": [
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 19,
      "chunk_id": "chunk_64",
      "distance": 0.9072,
      "text": ". The haptic system operates through a feedback loop. When a user applies force using a virtual instrument, sensors detect the input and transmit it to the simulation engine. T"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 22,
      "chunk_id": "chunk_79",
      "distance": 0.9445,
      "text": ". These systems can record and analyze various parameters such as accuracy, speed, tool handling, and procedural efficiency during simulated surgeries. Based on this data, 27"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 19,
      "chunk_id": "chunk_65",
      "distance": 0.95,
      "text": "corresponding response and sends signals to the haptic device, which produces the appropriate tactile sensation. This sensory feedback enhances realism and helps trainees devel"
    }
  ]
}
```

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